

Luis Alfredo Cuamatzi Flores

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SUMMARY

Software developer with experience across mobile apps, backend services, and AI integrations. I've worked in development and cross-functional teams in real production environments.

EXPERIENCE

Software & AI Developer

Net2Alliance

Jun 2025 – Present

Mexico City, Mexico

- Full-stack and AI development for telecom and CRM platforms.
- Built a FastAPI service for sentiment analysis and automated PDF summary generation from call transcripts.
- Developed a responsive WebSocket-based messaging client for WhatsApp with real-time notifications, multimedia support, and agent transfers.
- Implemented CRUD operations, role-based access control, audit logging, and reporting modules across admin dashboards.
- Participated in UI/UX redesigns with Figma and task tracking with Jira Kanban boards.

AI & Backend Engineer

L4L (3D Garment Reconstruction for Virtual Try-On)

Jan 2025 – Dec 2025

Tlaxcala, Mexico

- Built a distributed pipeline for 3D garment reconstruction from a single image.
- Developed a FastAPI orchestrator coordinating image capture, avatar confirmation, and GLB model generation.
- Model optimization: adapted SMPLer-X and Sewformer to run on local GPU-constrained environments, tuning inference pipelines for single-image input.
- Local Windows server using mayapy and Qualoth for physical cloth simulation and FBX-to-GLB conversion.

Android Developer

Instituto Politécnico Nacional – PIT

Oct 2024 – Aug 2025

Tlaxcala, Mexico

- Designed and built a native Android app to digitize attendance management for the IPN Institutional Tutoring Program.
- QR-based attendance scanning with duplicate detection within a 45-minute validation window.
- Weekly schedule builder with overlap detection for classrooms and tutors.
- Offline-first sync with Firestore and CSV export for institutional reporting.

EDUCATION

B.Eng. in Artificial Intelligence Engineering

Instituto Politécnico Nacional (IPN) – UPIIT

Aug 2021 – Jan 2026

Tlaxcala, Mexico

Leader Formation Program

Queen Mary University of London

Aug 2024

London, UK

PROJECTS

Kuromi Adventure

Mobile game in Unity

Jan 2026

- 2D/3D runner game with level progression, obstacles, and boss encounters.
- Built in Unity with C# scripting and touch controls for Android.
- **Tech:** Unity, C#, Android.

Lexfy

Android + AI project

Nov 2024

- Android app with OCR and AI image generation through a chat interface.
- Android client communicating via HTTP with a local Python/Flask backend running EasyOCR, GOT-OCR2_0, and Together AI.
- **Tech:** Kotlin, Jetpack Compose, Python, Flask, Firebase, EasyOCR, GOT-OCR2_0, Transformers.

SmartEMG Vision

Android prototype for assistive interaction

Nov 2024

- Mobile prototype combining EMG signal classification and object detection to suggest context-aware actions.

- Android client with Jetpack Compose and CameraX sending data via HTTP to local Python servers running YOLOv8 and TensorFlow/Keras.
- **Tech:** Kotlin, Jetpack Compose, CameraX, Python, Flask, TensorFlow, Keras, YOLOv8.

SmartCrops

Aug 2024

Mobile prototype for greenhouse monitoring

- Prototype for crop monitoring, greenhouse variable simulation, and AI-based chatbot assistance (TomaBot, powered by Gemini).
- Single-activity Jetpack Compose app with CameraX, OkHttp, and Coil.
- **Tech:** Kotlin, Jetpack Compose, Gemini API, CameraX, OkHttp, Coil.

EduStream API

Jun 2023

Web CMS for educational content

- Web system for uploading, categorizing, and streaming educational videos by subject.
- MVC backend with multimedia file management and synchronization between physical storage and the database.
- **Tech:** Node.js, Express, MySQL, Handlebars, Bootstrap, Multer.

TECHNICAL SKILLS

Languages: Kotlin, Python, JavaScript, TypeScript, PHP, C#, SQL

Mobile: Android native, Jetpack Compose, Material 3, CameraX, Room, WorkManager, OkHttp, Coil, ZXing, React Native, Expo

Backend: FastAPI, Flask, Node.js, Express, WebSockets, REST APIs, MVC architecture

AI / ML: TensorFlow, Keras, PyTorch, Transformers, EasyOCR, YOLOv8, SMPLer-X, Gemini API, Together AI

Databases & Cloud: Firestore, Firebase Realtime Database, MySQL, SQLite, Firebase Storage

Tools: Git, GitHub, Jira, Figma, Android Studio, VS Code, Postman

LANGUAGES

Spanish: Native

English: B2 (CEFR)